JAIPUR NATIONAL UNIVERSITY



Department of Biotechnology

TOPIC: *G-Protein in Signal Transduction*

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INTRODUCTION

G-Protein *or Guanine nucleotide-binding protein, plays an important role in Signal Transduction. It is the process of transferring signals from the outside of the cell to the inside, leading to a specific cellular response. These proteins acts as a switch that works back and forth between active and inactive states, it helps relay signals from receptors on the cell surface to internal pathways.*

Working of G-Proteins in signal transduction -:

- ***Inactive State:*** A G protein is inactive *when its alpha subunit is bound to guanosine diphosphate (GDP).*

- ***Active state:*** A G protein is active *when its alpha subunit binds to guanosine triphosphate (GTP).*

KEYWORDS-:

i. ***Heterotrimeric G proteins*** *– G-proteins subunits could be classified as, alpha (α), beta (β), and gamma (γ).*

ii. ***G Protein Coupled Receptor (GPCR)*** **-** *They are the largest and most diverse group of membrane receptors in eukaryotes. These cell surface receptors act like an inbox for messages in the form of light energy, peptides, lipids, sugars, and proteins.*

iii. ***GDP and GTP-*** *Guanosine diphosphate (GDP) and guanosine triphosphate (GTP) are nucleotides that are involved in a cycle that regulates biological functions:*

MECHANISM OF G-PROTEIN ACTIVATION AND ACTION

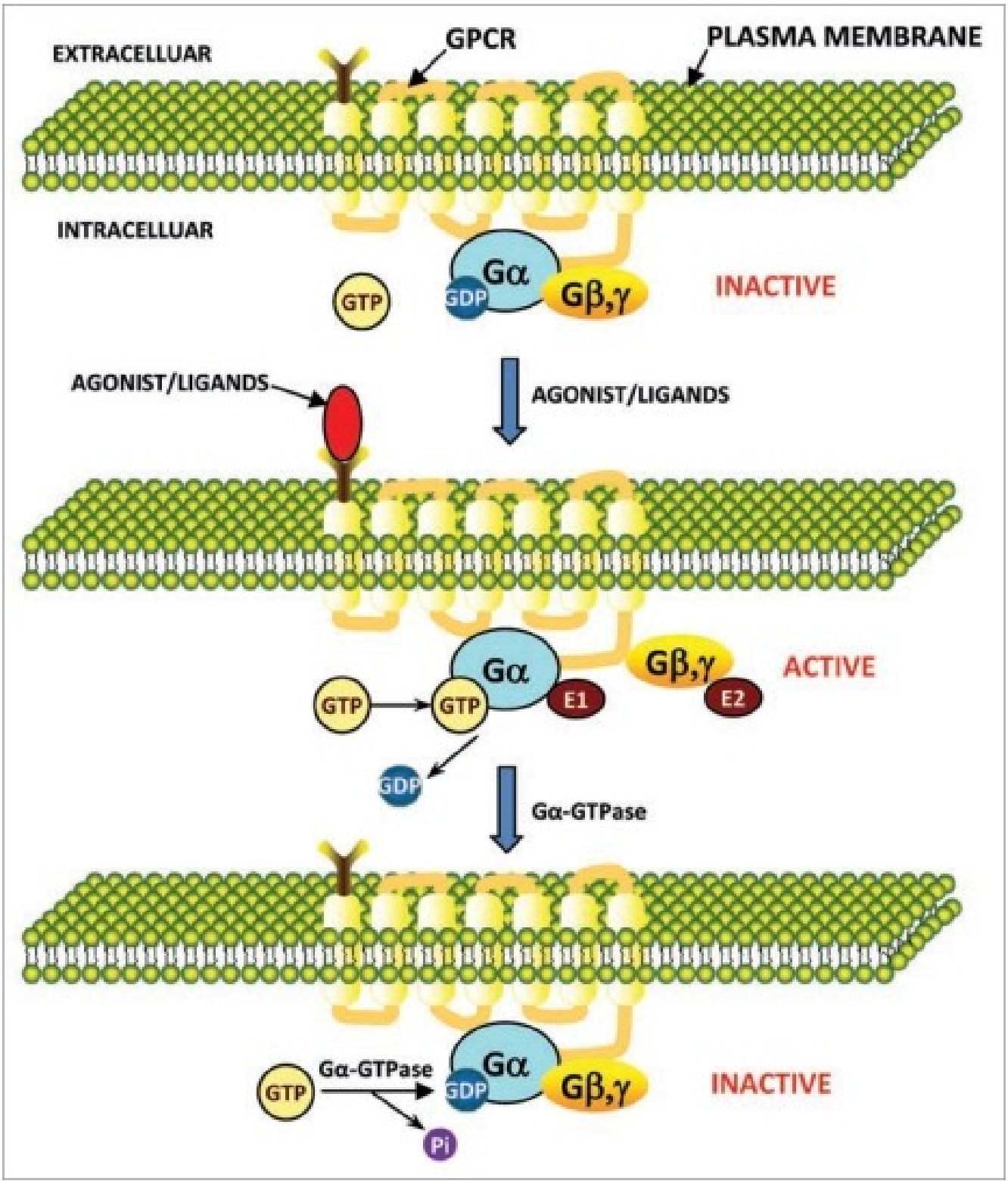
The regulatory cycle of G proteins i.e., activation/inactivation through GPCR . In the inactive state, Gα is bound to Gβγ dimer and GDP. G protein mediated signaling starts by binding of an agonist molecule that leads to activation of GPCR. GPCR is also a guanine nucleotide exchange factor that promotes the exchange of guanosine disphosphate (GDP)/guanosine triphosphate (GTP) associated with the Gα subunit. Therefore, the activated GPCR catalyzes exchange of GTP for GDP on the Gα subunit, as a result conformational changes takes place in the GPCR, which leads to dissociation of Gβγ dimer from Gα and thus activates multiple molecules of G proteins.

Activated Gα and Gβγ proteins in turn binds to various effectors and thereby switches it either on or off in different systems, and effectors continue to pass the signal to different kinds of second messengers. Here intrinsic GTPase activity of Gα comes into play, that leads to conversion of bound GTP into GDP and hence the inactivation of G proteins cascade. GTPase activity of the Gα subunits may also be regulated by regulators of G proteins signaling (RGS proteins) as well as effectors. Moreover, effector enzymes such as adenylyl cyclases may also regulate the activation of G proteins by receptors

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The activated Gα interacts and regulates many effector molecules such as calcium, potassium channels, adenylyl cyclase, phospholipase C (PLC), PLD and protein kinases. Initially it was hypothesized that the βγ dimer acts as negative regulator and can block activation of adenylyl cyclase by this mechanism.But subsequently this hypothesis was changed with the discovery that the βγ subunit could activate the muscarinic K+ channel and βγ subunits positively regulate effectors.Finally, the βγ subunit was shown to be a positive regulator of a large number of effectors in addition to the K+ channel, including adenylyl cyclase, phospholipase C-β (PLC-β), phospholipase A2 (PLA2), phosphoinositide 3-kinase (PI3-kinase), and β-adrenergic receptor kinase. Also, Gβγ can activate Gα subunit. It is now clear that many effectors are regulated both by α and βγ subunits.

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GPCR or G-Protein coupled receptors

GPCRs are the most common surface receptors, it is understandable that they are involved in a wide variety of physiological roles. GPCRs are involved in sight, taste, smell, behavior, mood, and immune system regulation. Even though the signaling molecules, types of GPCR, and mechanisms of action may differ for all these roles, they all involve certain extracellular signals that are converted into a cellular response.

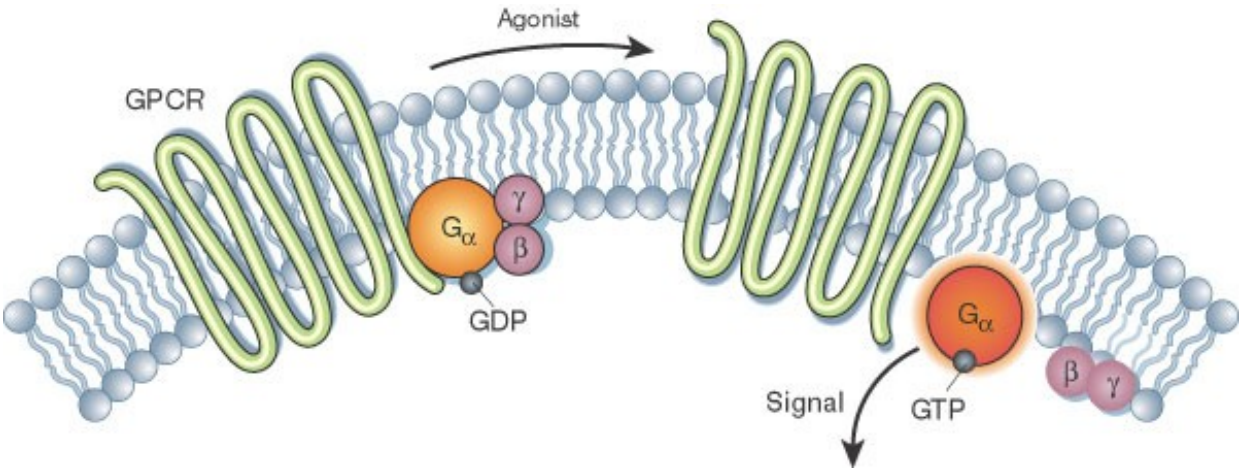
For example, mood regulation is affected by GPCRs because these receptors in the mammalian brain can bind to different neurotransmitters and create different physiological responses. Some molecules that can bind to GPCRs in the brain and lead to different moods are dopamine, serotonin, and GABA

MECHANISM

upon interaction with ligands or light activation, GPCRs transduce the signal into the cell by activating a cascade of biochemical events, which is initiated by catalyzing GDP-GTP exchange on heterotrimeric G proteins. Attenuation of GPCR signals in the presence of agonist/ligand is established by a coordinated series of events that are generally considered as three distinct events: receptor desensitization, sequestration/internalization, and downregulation. Desensitization begins within seconds of agonist exposure and is initiated by phosphorylation of the receptor.

ISSUES WITH GPCR

GPCRs are the largest class of membrane-bound receptors and one of the most prevalent gene families. These receptors are activated by various stimuli such as hormones, neurotransmitters, chemokines, odorants, and others. GPCRs influence broad physiological processes such as neurotransmission, cellular metabolism, secretion, cell growth, and immune responses.



STRUCTURE

GPCRs bind a tremendous variety of signaling molecules, yet they share a common architecture that has been conserved over the course of evolution. Many present-day eukaryotes — including animals, plants, fungi, and protozoa — rely on these receptors to receive information from their environment. For example, simple eukaryotes such as yeast have GPCRs that sense glucose and mating factors. Not surprisingly, GPCRs are involved in considerably more functions in multicellular organisms. Humans alone have nearly 1,000 different GPCRs, and each one is highly specific to a particular signal.

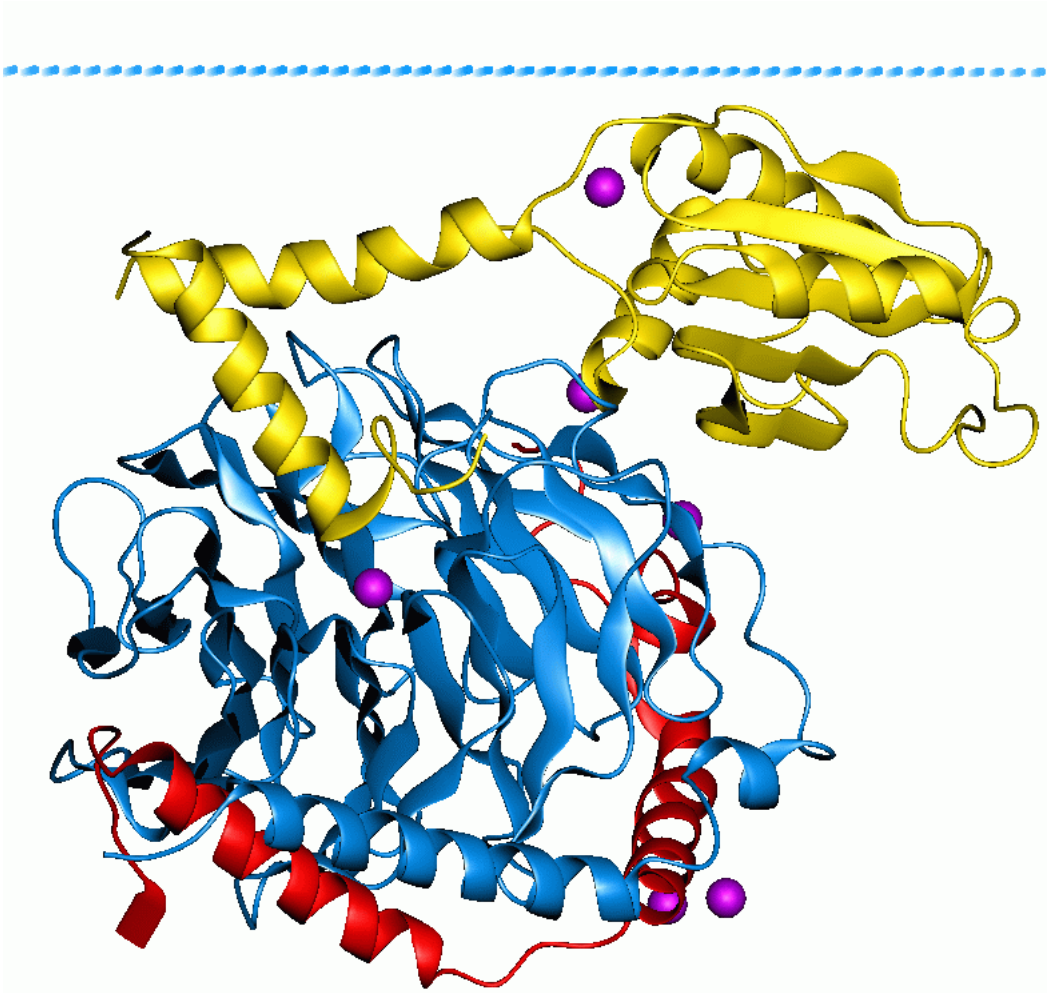
GPCRs consist of a single polypeptide that is folded into a globular shape and embedded in a cell's plasma membrane. Seven segments of this molecule span the entire width of the membrane — explaining why GPCRs are sometimes called ***seven-transmembrane receptors*** *— and the intervening portions loop both inside and outside the cell. The extracellular loops form part of the pockets at which signaling molecules bind to the GPCR.*

G-Protein

***G proteins**, also known as **guanine nucleotide-binding proteins**, are a family of proteins that act as molecular switches inside cells, and are involved in transmitting signals from a variety of stimuli outside a cell to its interior. Their activity is regulated by factors that control their ability to bind to and hydrolyze guanosine triphosphate (GTP) to guanosine diphosphate (GDP). When they are bound to GTP, they are 'on', and, when they are bound to GDP, they are 'off'. G proteins belong to the larger group of enzymes called GTPases.*

There are two classes of G proteins.

HETEROTRIMIC and MONOMERIC



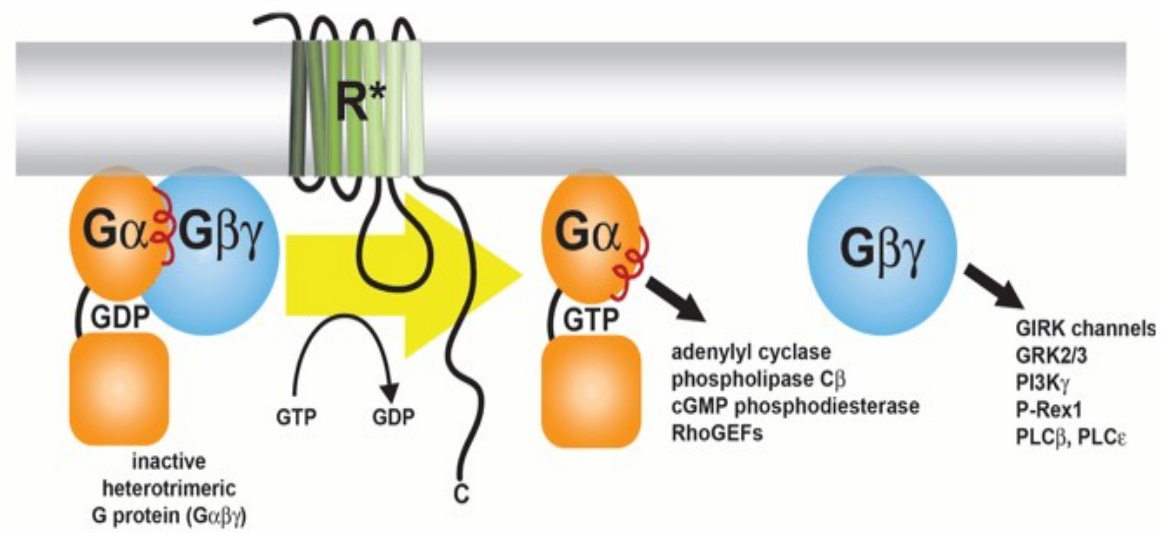
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HETEROTRIMERIC G PROTEINS

Heterotrimeric G proteins are a family of membrane proteins that are made up of three subunits, alpha (α), beta (β), and gamma (γ). They are involved in many cellular processes, including:

- *Signal transduction: Heterotrimeric G proteins act as molecular switches in signal transduction pathways, transducing signals from the plasma membrane to intracellular effectors.*
- *Protein synthesis: Heterotrimeric G proteins are involved in protein synthesis.*
- *Vesicle movement: Heterotrimeric G proteins are involved in vesicle movement inside cells.*
- *Cell growth regulation: Heterotrimeric G proteins are involved in cell growth regulation.*

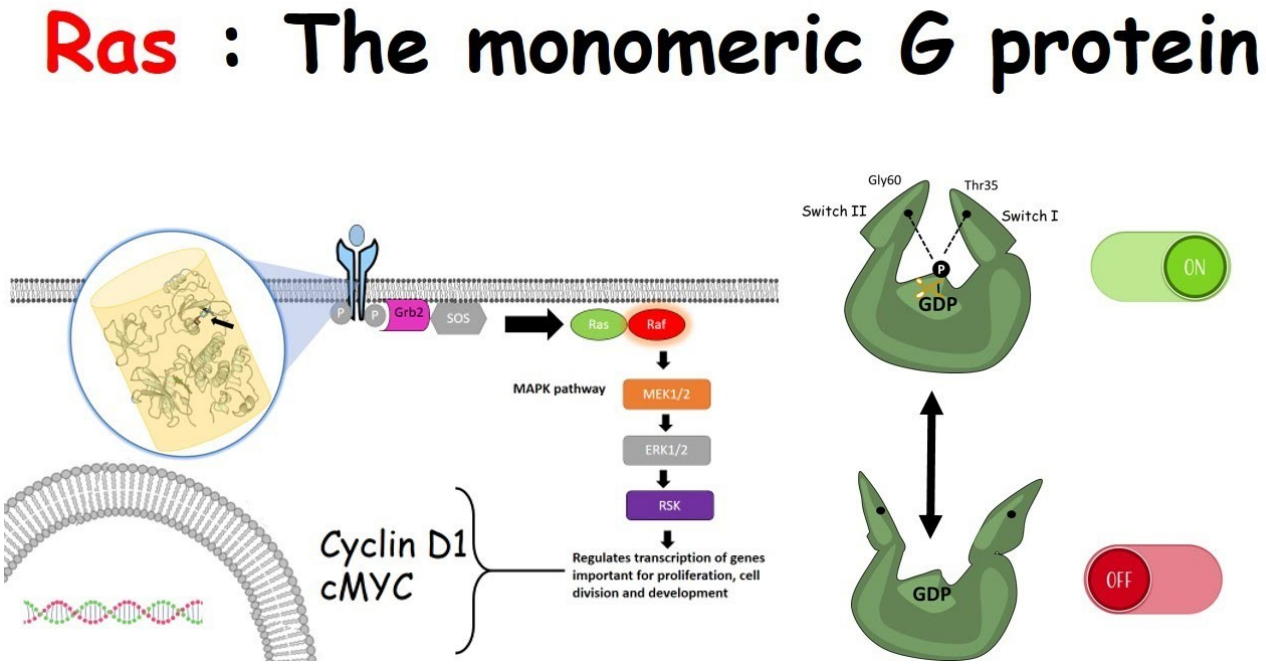
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MONOMERIC G-PROTEIN

Monomeric G proteins, also known as small G proteins or small GTPases, are proteins that act as molecular switches to relay signals from cell surface receptors to intracellular targets. They are involved in many cellular processes, including: Nuclear localization of proteins, Endocytosis, Cell adhesion and migration, and Changes to the actin cytoskeleton

Monomeric G proteins are activated by extracellular stimuli and cycle between GDP-bound (inactive) and GTP-bound (active) states. They are structurally related to the α subunit of trimeric G proteins



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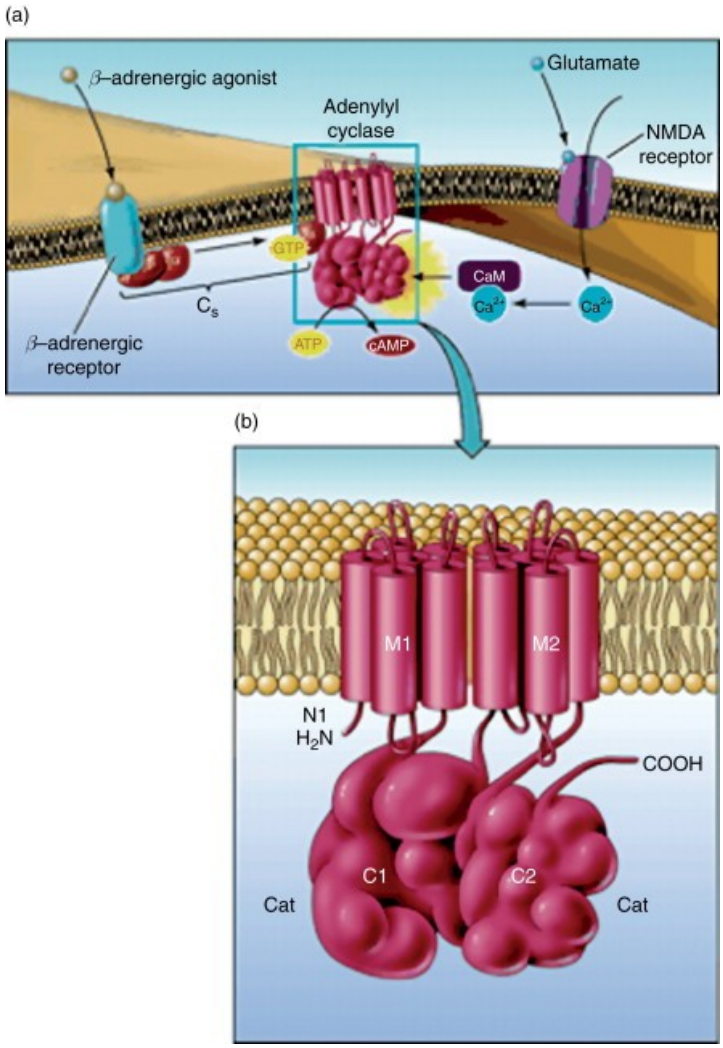
Effector Proteins Activated by G-Proteins

Adenylyl Cyclase (AC):*Produces cAMP from ATP (Gs pathway).*

Phospholipase C (PLC): *Hydrolyzes phosphatidylinositol bisphosphate (PIP2) to IP3 and DAG (Gq pathway).*

Ion Channels: *Gβγ subunits can regulate ion channels, such as G-protein gated inward rectifying potassium (GIRK) channels or calcium channels.*

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SECOND MESSENGERS

Second messengers are small molecules that are released inside a cell as part of a signal transduction cascade that is triggered by activated cell surface receptors. Some examples of second messengers in G protein signaling include:

Cyclic AMP (cAMP)

A common second messenger that is synthesized from ATP by adenylyl cyclase, an enzyme that is activated by the GTP-bound alpha subunit of a G protein. cAMP is involved in responses to hormones, sensory input, and nerve transmission.

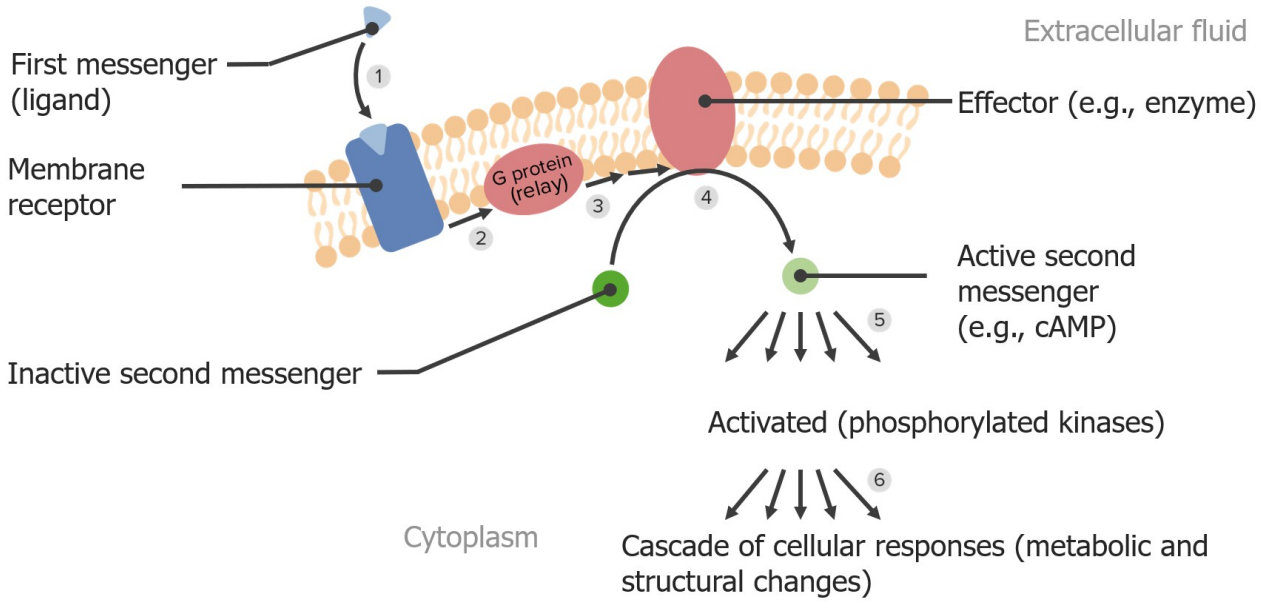
Inositol 1, 4, 5-triphosphate (IP3)

A second messenger that is synthesized from phosphatidyl inositol by phospholipase C, an enzyme that is activated by G proteins. IP3 acts on IP3 receptors to release calcium from the endoplasmic reticulum (ER)

Diacylglycerol (DAG)

A second messenger that is synthesized from phosphatidyl inositol by phospholipase C, an enzyme that is activated by G proteins. DAG activates protein kinase C (PKC).

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DYSREGULATION OF G PROTEIN

DISEASES

Dysregulation of G protein signaling can lead to many diseases, including:

- Cancer: G protein-coupled receptors (GPCRs) are involved in cell division, and genetic abnormalities in GPCR signaling pathways can lead to cancer.
- Cardiovascular problems: Abnormal GPCR signaling can lead to cardiac problems.
- Asthma: Abnormal GPCR signaling can lead to asthma.
- Retinitis pigmentosa: Mutations in GPCRs can cause retinitis pigmentosa.
- Hypo- and hyperthyroidism: Mutations in GPCRs can cause hypo- and hyperthyroidism.
- Nephrogenic diabetes insipidus: Mutations in GPCRs can cause nephrogenic diabetes insipidus.
- Fertility disorders: Mutations in GPCRs can cause fertility disorders.

CONCLUSION

In conclusion, G-Protein play a major role in signal transduction, acting as a key intermediaries that relays signal from activated GPCRs to intracellular effectors. Through the activation of various G-protein subunits (Gαs, Gαi, Gαq, etc.), they regulate a wide range of cellular processes by modulating second messengers like cAMP, IP3, DAG, and calcium. This intricate system allows cells to respond dynamically to extracellular signals, controlling functions such as metabolism, growth, sensory perception, and hormone regulation. The precise regulation of G-protein signaling is critical for maintaining cellular homeostasis, and its dysregulation can lead to diseases like cancer, metabolic disorders, and neurological conditions. Understanding G-protein mechanisms provides crucial insights into therapeutic interventions targeting GPCR pathways.

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